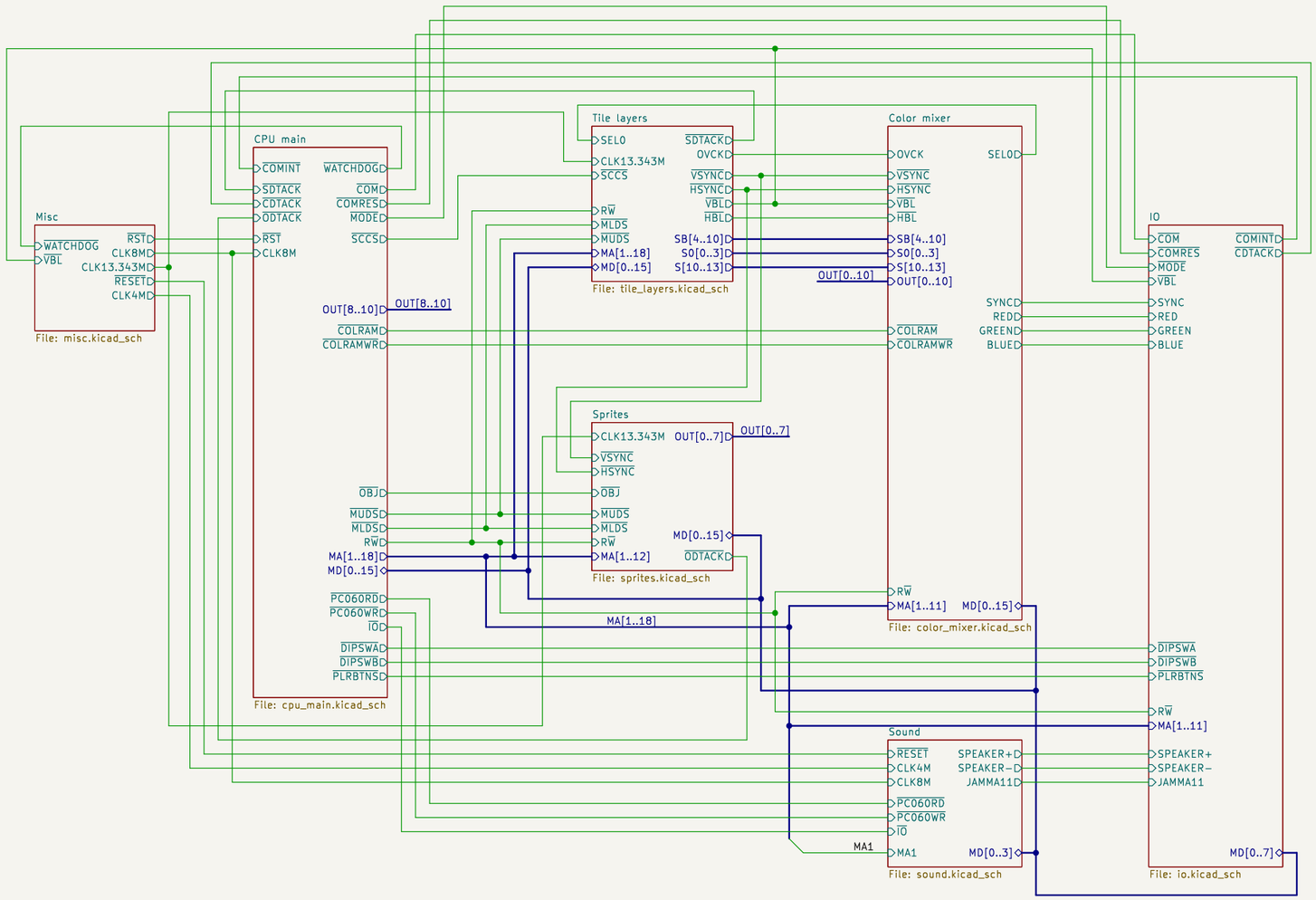


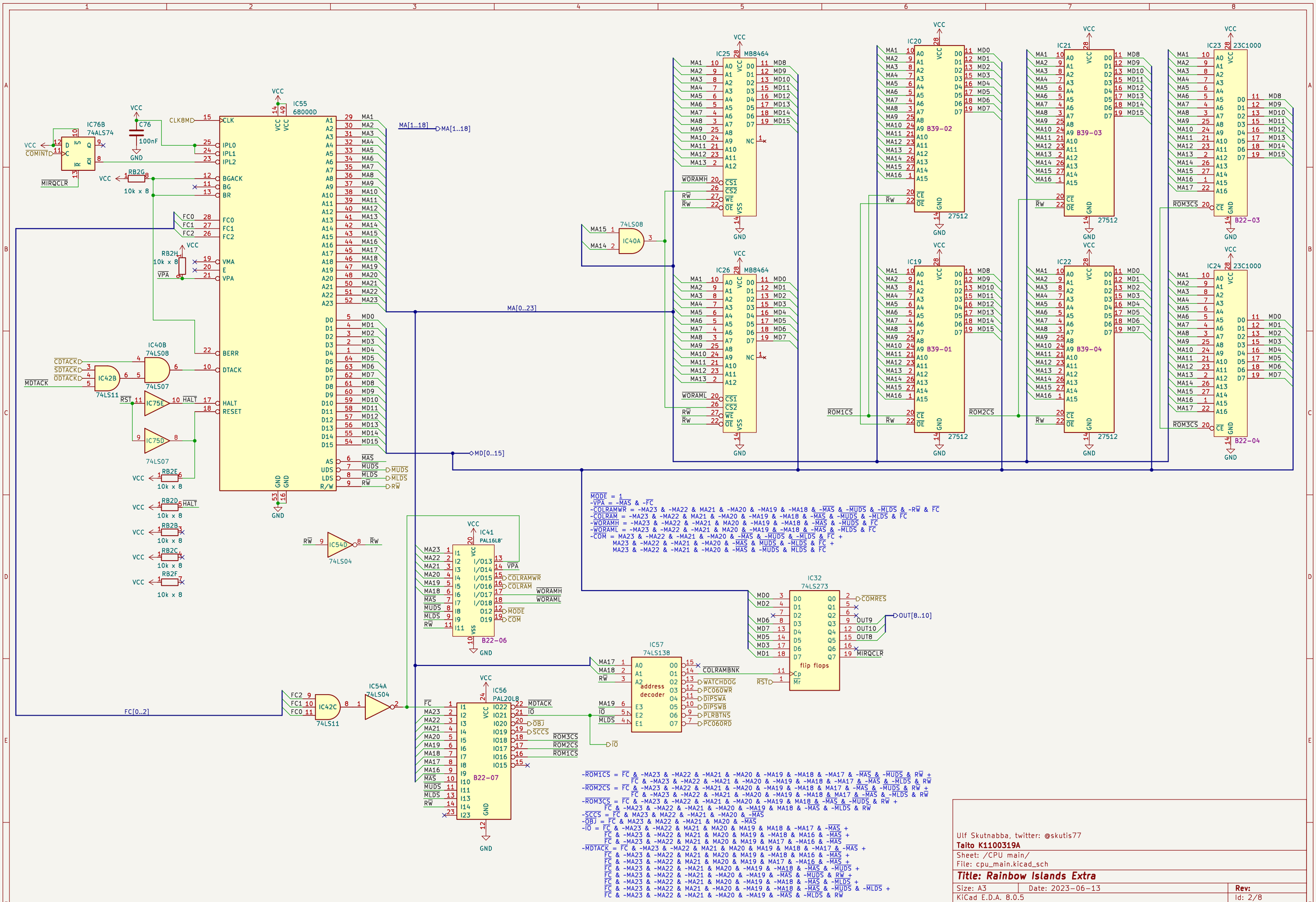
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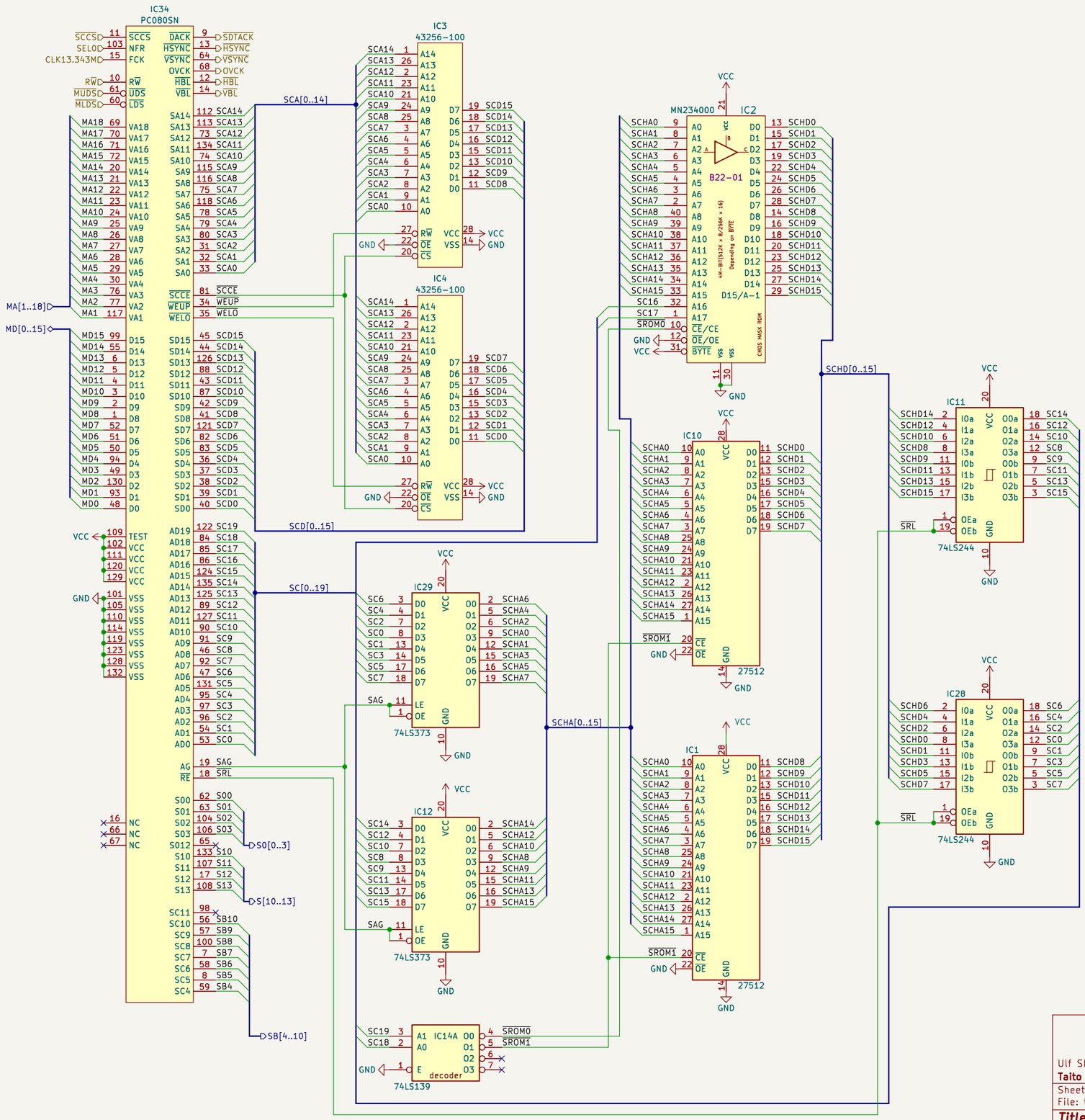
Taito K1100319A

Sheet: /
File: risle.kicad_sch

Title: Rainbow Islands Extra

Size: A3	Date: 2023-06-13	**Rev:**
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IC55
68000D
IC76B
74ALS74
C76
100nF
RB2G
10k x 8
RB2H
10k x 8
RB2E
10k x 8
RB2D
10k x 8
RB2B
10k x 8
RB2C
10k x 8
RB2F
10k x 8
IC40B
74LS08
IC42B
74LS11
IC75E
IC75D
74LS07
IC54D
74LS04
IC54A
74LS04
IC42C
74LS11
IC40A
74LS08
IC25 MBB464
IC26 MBB464
IC20 B39-02
IC19 B39-01
IC21 B39-03
IC22 B39-04
IC23 23C1000
IC24 23C1000
B22-03
B22-04
27512
IC41 PAL16L8'
B22-06
IC56 PAL20L8
B22-07
IC57 74LS138
address decoder
IC32 74LS273
flip flops
CLK8MD
COMINTD
MIRQCLR
MA[1..18]
MA[0..23]
MD[0..15]
FC[0..2]
VPA
RW
MDTACK
RST
HALT
BERR
DTACK
BGACK
BG
BR
VMA
E
FC0
FC1
FC2
IPL0
IPL1
IPL2
AS
UDS
LDS
R/W
MAS
MUDS
MLDS
ROM1CS
ROM2CS
ROM3CS
COLRAMBNK
WATCHDOG
PC060WR
DIPSWA
DIPSWB
PLRBTNS
PC060RD
COMRES
OUT[8..10]
OUT9
OUT10
OUT8
MIRQCLR
WORAMH
WORAML
COLRAMWR
COLRAM
MODE
COM
OBJ
SCCS
IO
MODE = 1
-VPA = ~MAS & ~FC
-COLRAMWR = ~MA23 & ~MA22 & MA21 & ~MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS & ~MLDS & ~RW & FC
-COLRAM = ~MA23 & ~MA22 & MA21 & ~MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS & ~MLDS & FC
-WORAMH = ~MA23 & ~MA22 & ~MA21 & MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS & FC
-WORAML = ~MA23 & ~MA22 & ~MA21 & MA20 & ~MA19 & ~MA18 & ~MAS & ~MLDS & FC
-COM = MA23 & ~MA22 & ~MA21 & ~MA20 & ~MAS & ~MUDS & ~MLDS & FC +
MA23 & ~MA22 & ~MA21 & ~MA20 & ~MAS & MUDS & ~MLDS & FC +
MA23 & ~MA22 & ~MA21 & ~MA20 & ~MAS & ~MUDS & MLDS & FC
-ROM1CS = FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MA18 & ~MA17 & ~MAS & ~MUDS & RW +
FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MA18 & ~MA17 & ~MAS & ~MLDS & RW
-ROM2CS = FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MA18 & MA17 & ~MAS & ~MUDS & RW +
FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MA18 & MA17 & ~MAS & ~MLDS & RW
-ROM3CS = FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & MA18 & ~MAS & ~MUDS & RW +
FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & MA18 & ~MAS & ~MLDS & RW
-SCCS = FC & MA23 & MA22 & ~MA21 & ~MA20 & ~MAS
-OBJ = FC & MA23 & MA22 & ~MA21 & MA20 & ~MAS
-IO = FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & MA18 & ~MA17 & ~MAS +
FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & ~MA18 & MA16 & ~MAS +
FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & MA17 & ~MA16 & ~MAS
-MDTACK = FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & MA18 & ~MA17 & ~MAS +
FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & ~MA18 & MA16 & ~MAS +
FC & ~MA23 & ~MA22 & MA21 & MA20 & MA19 & MA17 & ~MA16 & ~MAS +
FC & ~MA23 & ~MA22 & ~MA21 & MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS +
FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MAS & ~MUDS & RW +
FC & ~MA23 & ~MA22 & ~MA21 & MA20 & ~MA19 & ~MA18 & ~MAS & ~MLDS +
FC & ~MA23 & ~MA22 & MA21 & ~MA20 & ~MA19 & ~MA18 & ~MAS & ~MUDS & ~MLDS +
FC & ~MA23 & ~MA22 & ~MA21 & ~MA20 & ~MA19 & ~MAS & ~MLDS & RW
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Sheet: /CPU main/
File: cpu_main.kicad_sch
Title: Rainbow Islands Extra
Size: A3
Date: 2023-06-13
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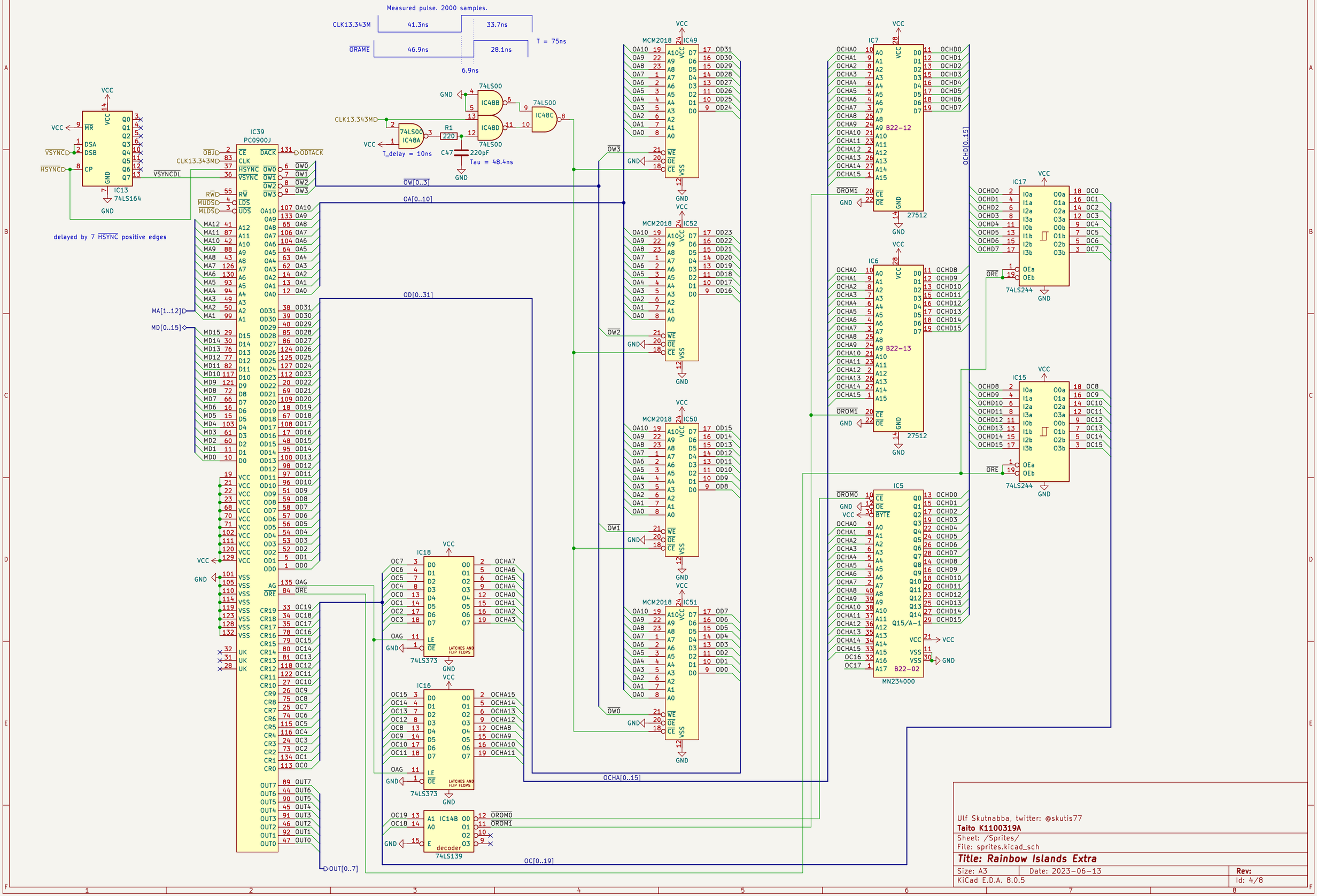
Taito K1100319A

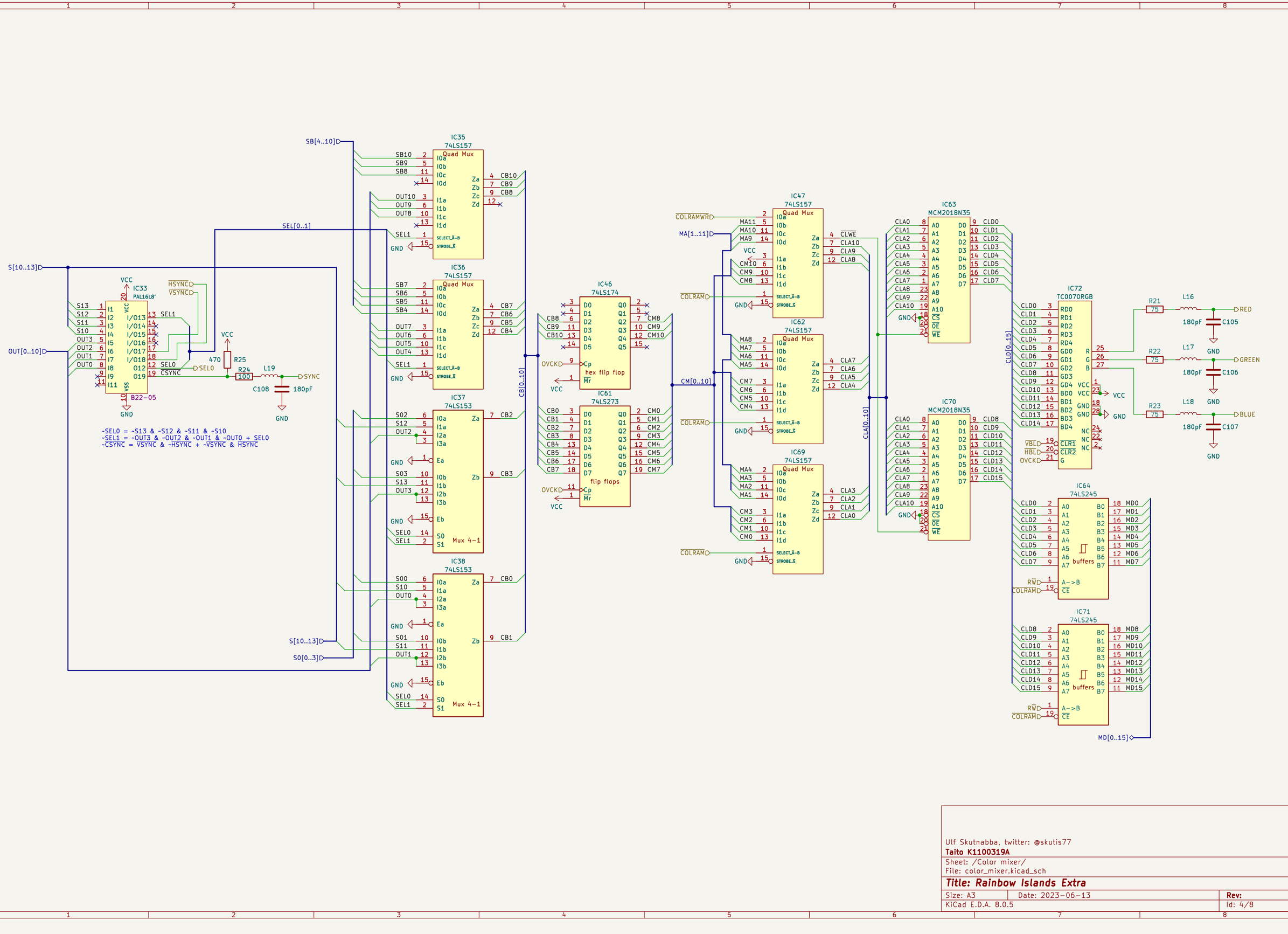
Sheet: /Tile layers/

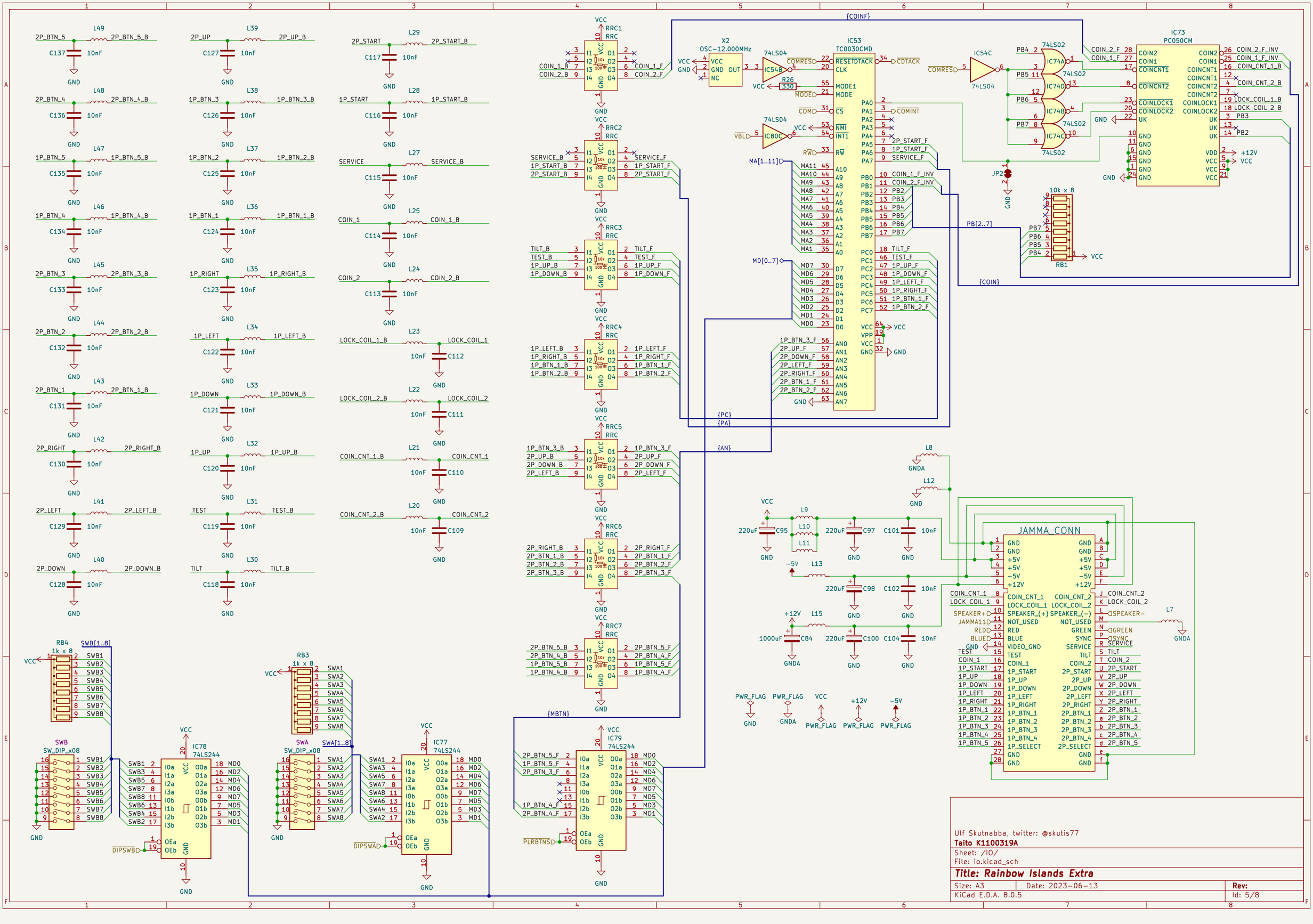
File: tile_layers.kicad_sch

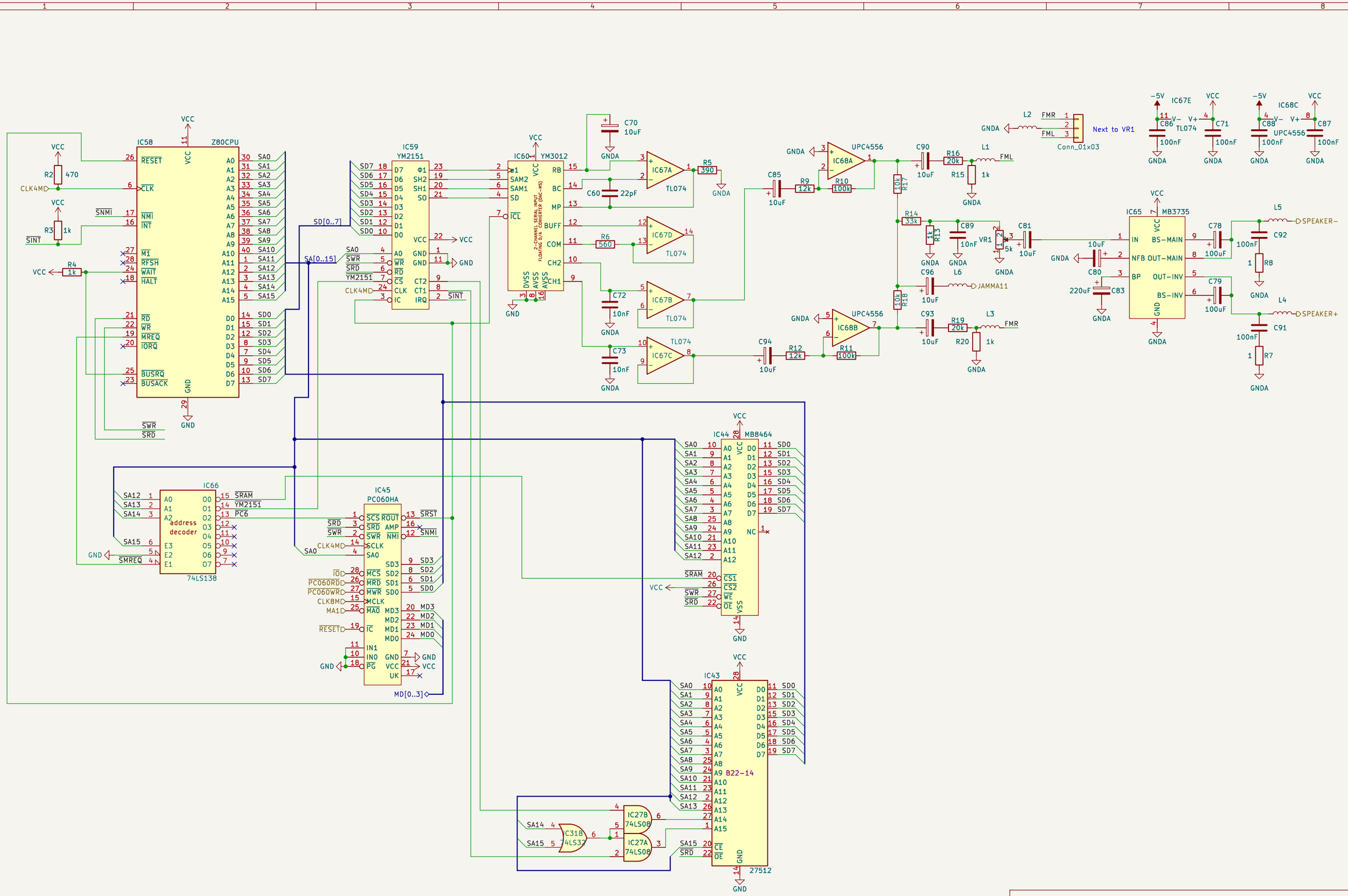
Title: Rainbow Islands Extra

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Measured pulse. 2000 samples.
CLK13.343M
41.3ns
33.7ns
T = 75ns
ORAME
46.9ns
28.1ns
6.9ns
74LS00
IC48B
IC48C
IC48D
IC48A
R1
220
C47
220pF
Tau = 48.4ns
T_delay = 10ns
IC13
74LS164
delayed by 7 HSYNC positive edges
IC39
PC0900J
VSYNCDL
OW[0..3]
OA[0..10]
OD[0..31]
MA[1..12]
MD[0..15]
OUT[0..7]
OCHA[0..15]
OC[0..19]
MCM2018
IC49
IC52
IC50
IC51
IC18
IC16
74LS373
IC14B
decoder
74LS139
IC7
IC6
IC5
IC17
IC15
27512
74LS244
MN234000
OCHD[0..15]
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Taito K1100319A
Sheet: /Sprites/
File: sprites.kicad_sch
Title: Rainbow Islands Extra
Size: A3
Date: 2023-06-13
KiCad E.D.A. 8.0.5
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SB[4..10]
SEL[0..1]
S[10..13]
OUT[0..10]
S0[0..3]
CB[0..10]
CM[0..10]
CLA[0..10]
MD[0..15]
IC35
74LS157
Quad Mux
IC36
74LS157
Quad Mux
IC37
74LS153
Mux 4-1
IC38
74LS153
Mux 4-1
IC33
PAL16L8'
B22-05
IC46
74LS174
hex flip flop
IC61
74LS273
flip flops
IC47
74LS157
Quad Mux
IC62
74LS157
Quad Mux
IC69
74LS157
Quad Mux
IC63
MCM2018N35
IC70
MCM2018N35
IC72
TC0070RGB
IC64
74LS245
buffers
IC71
74LS245
buffers
~SEL0 = ~S13 & ~S12 & ~S11 & ~S10
~SEL1 = ~OUT3 & ~OUT2 & ~OUT1 & ~OUT0 + SEL0
~CSYNC = VSYNC & ~HSYNC + ~VSYNC & HSYNC
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Taito K1100319A
Sheet: /Color mixer/
File: color_mixer.kicad_sch
Title: Rainbow Islands Extra
Size: A3
Date: 2023-06-13
KiCad E.D.A. 8.0.5
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2P_BTN_5
L49
2P_BTN_5_B
C137
10nF
GND
2P_BTN_4
L48
2P_BTN_4_B
C136
1P_BTN_5
L47
1P_BTN_5_B
C135
1P_BTN_4
L46
1P_BTN_4_B
C134
2P_BTN_3
L45
2P_BTN_3_B
C133
2P_BTN_2
L44
2P_BTN_2_B
C132
2P_BTN_1
L43
2P_BTN_1_B
C131
2P_RIGHT
L42
2P_RIGHT_B
C130
2P_LEFT
L41
2P_LEFT_B
C129
2P_DOWN
L40
2P_DOWN_B
C128
2P_UP
L39
2P_UP_B
C127
1P_BTN_3
L38
1P_BTN_3_B
C126
1P_BTN_2
L37
1P_BTN_2_B
C125
1P_BTN_1
L36
1P_BTN_1_B
C124
1P_RIGHT
L35
1P_RIGHT_B
C123
1P_LEFT
L34
1P_LEFT_B
C122
1P_DOWN
L33
1P_DOWN_B
C121
1P_UP
L32
1P_UP_B
C120
TEST
L31
TEST_B
C119
TILT
L30
TILT_B
C118
2P_START
L29
2P_START_B
C117
1P_START
L28
1P_START_B
C116
SERVICE
L27
SERVICE_B
C115
COIN_1
L25
COIN_1_B
C114
COIN_2
L24
COIN_2_B
C113
LOCK_COIL_1_B
L23
LOCK_COIL_1
C112
LOCK_COIL_2_B
L22
LOCK_COIL_2
C111
COIN_CNT_1_B
L21
COIN_CNT_1
C110
COIN_CNT_2_B
L20
COIN_CNT_2
C109
RRC1
RRC2
RRC3
RRC4
RRC5
RRC6
RRC7
RRC
10k
100R
COIN_1_B
COIN_2_B
COIN_1_F
COIN_2_F
SERVICE_B
1P_START_B
2P_START_B
SERVICE_F
1P_START_F
2P_START_F
TILT_B
TEST_B
1P_UP_B
1P_DOWN_B
TILT_F
TEST_F
1P_UP_F
1P_DOWN_F
1P_LEFT_B
1P_RIGHT_B
1P_BTN_1_B
1P_BTN_2_B
1P_LEFT_F
1P_RIGHT_F
1P_BTN_1_F
1P_BTN_2_F
1P_BTN_3_B
2P_UP_B
2P_DOWN_B
2P_LEFT_B
1P_BTN_3_F
2P_UP_F
2P_DOWN_F
2P_LEFT_F
2P_RIGHT_B
2P_BTN_1_B
2P_BTN_2_B
2P_BTN_3_B
2P_RIGHT_F
2P_BTN_1_F
2P_BTN_2_F
2P_BTN_3_F
2P_BTN_5_B
2P_BTN_4_B
1P_BTN_5_B
1P_BTN_4_B
2P_BTN_5_F
2P_BTN_4_F
1P_BTN_5_F
1P_BTN_4_F
X2
OSC-12.000MHz
IC54B
74LS04
IC54C
IC8B0C
IC53
TC0030CMD
RESETDTACK
CLK
MODE1
MODE
CS
NMI
INT1
RW
MA[1..11]
MD[0..7]
CDTACK
COMRES
COMINT
VBLL
R26
330
JP2
{COINF}
{PC}
{PA}
{AN}
{MBTN}
{COIN}
PB[2..7]
IC73
PC050CM
COIN2
COIN1
COINCNT1
COINCNT2
COINLOCK1
COINLOCK2
UK
VDD
+12V
PB3
PB2
10k x 8
RB1
IC74A
IC74B
IC74C
IC74D
74LS02
RB4
1k x 8
SWB[1..8]
SWB
SW_DIP_x08
IC78
74LS244
DIPSWB
RB3
SWA
SWA[1..8]
IC77
DIPSWA
IC79
PLRBTNS
L8
GNDA
L12
L9
L10
L11
220uF
C95
C97
C101
C98
C102
C100
C104
C84
1000uF
L13
L15
L7
SPEAKER+
SPEAKER-
JAMMA_CONN
JAMMA11
RED
BLUE
GREEN
SYNC
SERVICE
PWR_FLAG
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Taito K1100319A
Sheet: /IO/
File: io.kicad_sch
Title: Rainbow Islands Extra
Size: A3
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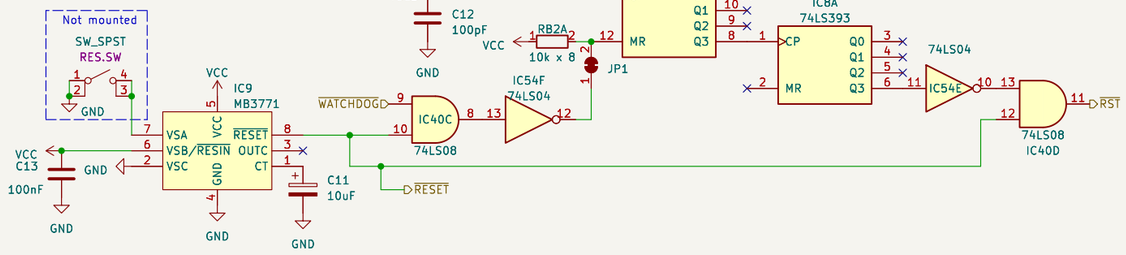
Sheet: /Sound/

File: sound.kicad_sch

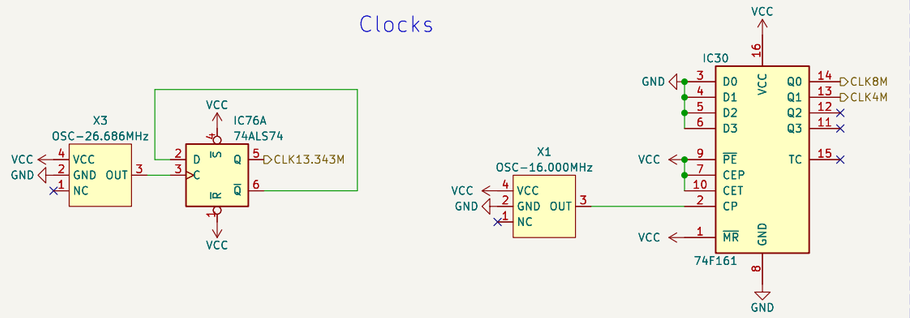
Title: Rainbow Islands Extra

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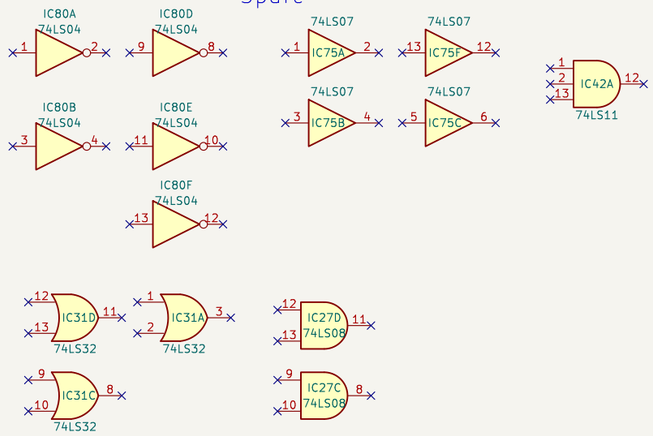
Watchdog and reset



Clocks



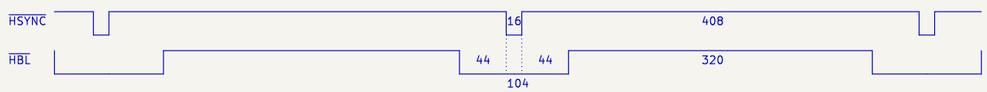
Spare



Horizontal and vertical synch timing diagrams

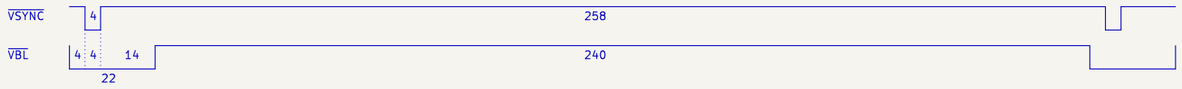
The pixel clock is derived from the 26.686MHz oscillator.
Pixel clock OVCK: f = 26.686MHz / 4 = 6.6715MHz

The numbers in the HSYNC and HBL diagram are pixel clock cycles.
All edges are synchronised to the rising edge of OVCK.



HSYNC and HBL
Frequency f = 6.6715MHz / 424 = 15.734670kHz.
Period T = 1/f = 63.5539us.

The numbers in the VSYNC and VBL diagram are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



VSYNC and VBL:
Frequency f = 15.734670kHz / 262 = 60.05599Hz
Period T = 1/f = 1 / 15.734670kHz = 16.6511ms

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Sheet: /Misc/
File: misc.kicad_sch

Title: Rainbow Islands Extra

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